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### Power supply system and power supply unit

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#### Abstract

Provided is a power supply system in which two or more power supply units are optionally connected and used in any operation mode. A power supply system **100** includes two or more power supply units **10** to **13**, the power supply units **10** to **13** cooperating to supply DC power to a shared load **22**, the power supply units **10** to **13** including a power supply main circuit **30** that converts AC power inputted from the outside into DC power; a control unit **36** that controls the power supply main circuit **30**; a pair of synchronous terminals **Z1**, **Z2** respectively connected to a pair of synchronous signal lines **ZCL**; a transmission unit **35** that outputs an error signal **ER** by establishing conduction between the pair of synchronous signal lines **ZCL** upon detection of an abnormality; and a reception unit **35** that detects a conductive state between the pair of synchronous signal lines **ZCL** and that receives the error signal **ER**, wherein the power supply main circuit **30** stops outputting based on timing of receiving the error signal **ER**.

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| <b>Inventors:</b>            | <b>Makitani; Atsushi (Osaka, JP), Uenoyama; Taro (Osaka, JP)</b> |
| <b>Applicant:</b>            | <b>Sansha Electric Manufacturing Co., Ltd. (Osaka, JP)</b>       |
| <b>Family ID:</b>            | <b>81213079</b>                                                  |
| <b>Assignee:</b>             | <b>SANSHA ELECTRIC MANUFACTURING CO., LTD. (Osaka, JP)</b>       |
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*Primary Examiner:* Berhane; Adolf D

*Assistant Examiner:* Demisse; Afework S

*Attorney, Agent or Firm:* Kilyk & Bowersox, P.L.L.C.

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## **Background/Summary**

(1) This application is a National Stage Application of PCT/JP2021/02749, filed Jan. 27, 2021.  
TECHNICAL FIELD

(2) The present invention relates to a power supply system and a power supply unit, and more particularly to a power supply system that includes two or more power supply units and in which the power supply units cooperate to supply DC power to a shared load.

### **BACKGROUND ART**

(3) A power supply system that includes two or more power supply units and that supplies DC power from each power supply unit to a common load is known (for example, Patent Literature 1 and 2). In this type of power supply system, a power supply apparatus having various characteristics can be realized by connecting two or more power supply units. For example, in an operation test of an electric vehicle or a power conditioner, or the like, such a power supply system is used as a simulated power supply that simulates a battery, a solar panel, or the like.

(4) In this kind of power supply system, in a case where an abnormality such as failure occurs in any of the power supply units, an excessive load is applied to the other power supply units due to a reduction in the output of the power supply unit, and there is a risk of damage to the normal power supply units. Therefore, it is necessary to quickly transmit the occurrence of abnormality to the other power supply units and stop the outputs of all the power supply units. In addition, there is a problem that the operation becomes unstable when a deviation occurs in power supply unit stoppage operations.

### **CITATION LIST**

(5) Patent Literature 1: JP 2009-148032 A Patent Literature 2: JP 2014-147196 A

### **SUMMARY OF INVENTION**

#### **Technical Problem**

(6) The present invention was conceived in view of the above circumstances, and an object of the present invention is to provide a power supply system capable of quickly transmitting an abnormality occurring in any of two or more power supply units constituting the power supply system, between the power supply units. Another object is to provide a power supply system that quickly stops outputting when an abnormality occurs in any of the power supply units.

#### **Solution to Problem**

(7) A power supply system according to a first embodiment of the present invention is a power supply system including two or more power supply units, the power supply units cooperating to supply DC power to a shared load, the power supply units including a power supply main circuit

that converts AC power inputted from the outside into DC power; a control unit that controls the power supply main circuit; a pair of synchronous terminals respectively connected to a pair of synchronous signal lines; a transmission unit that outputs an error signal by establishing conduction between the pair of synchronous signal lines upon detection of an abnormality; and a reception unit that detects a conductive state between the pair of synchronous signal lines and that receives the error signal, wherein the power supply main circuit stops outputting based on timing of receiving the error signal.

(8) By adopting such a configuration, when an abnormality occurs in any of the power supply units, the occurrence of the abnormality can be quickly transmitted to all the other power supply units. In addition, the operation to stop the output of each power supply unit can be synchronized, and the operation of the power supply system when the output is stopped can be stabilized.

(9) The power supply system according to the second embodiment of the present invention is configured such that, in addition to the above configuration, the transmission unit and the reception unit each include a photocoupler, and the pair of synchronous signal lines is isolated from the power supply units by the photocoupler.

(10) A power supply system according to a third embodiment of the present invention is configured such that, in addition to the above configuration, the power supply main circuit starts outputting based on an edge of the error signal.

(11) A power supply unit according to a fourth embodiment of the present invention is a power supply unit that is connected to another power supply unit and that cooperates with the another power supply unit to supply DC power to a shared load, the power supply unit including a power supply main circuit that converts AC power inputted from the outside into DC power; a control unit that controls the power supply main circuit; a pair of synchronous terminals respectively connected to a pair of synchronous signal lines; a transmission unit that outputs an error signal by establishing conduction between the pair of synchronous signal lines upon detection of an abnormality; and a reception unit that detects a conductive state between the pair of synchronous signal lines and that receives the error signal, wherein the power supply main circuit stops outputting based on timing of receiving the error signal.

#### Advantageous Effects of Invention

(12) With the present invention, it is possible to provide a power supply system capable of quickly transmitting an abnormality occurring in any of two or more power supply units constituting the power supply system, between the power supply units. A power supply system that quickly stops outputting when an abnormality occurs in any of the power supply units can also be provided.

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## Description

### BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a diagram illustrating an example of a schematic configuration of a power supply system **100** according to a first embodiment of the present invention.

(2) FIG. 2 is a diagram showing an example of a detailed configuration of the power supply units **10** to **13**.

(3) FIG. 3 is a diagram showing an example of a detailed configuration of the synchronous processing unit **35**.

(4) FIG. 4 is a timing chart showing an example of an output start operation of the power supply system **100**.

(5) FIG. 5 is a flowchart showing an example of the operation of the master unit MU.

(6) FIG. 6 is a flowchart showing an example of the operation of the slave units SU.

(7) FIG. 7 is a timing chart showing an example of the operation of the power supply system **100** according to a second embodiment of the present invention.

(8) FIG. 8 is a timing chart showing another example of the operation of the power supply system **100** according to the second embodiment of the present invention.

## DESCRIPTION OF EMBODIMENTS

### First Embodiment

(9) (1) Outline of Power Supply System **100**

(10) FIG. 1 is a diagram illustrating an example of a schematic configuration of a power supply system **100** according to a first embodiment of the present invention. The power supply system **100** is a power supply apparatus that converts AC power **21** supplied from the power grid into the desired DC power and supplies the DC power to a load **22**.

(11) The power supply system **100** includes two or more power supply units **10** to **13** connected to the load **22** via an output line PoL. Each of the power supply units **10** to **13** converts AC power supplied from the outside into DC power, based on a control parameter Pr from a control terminal **20**, and supplies the DC power to the load **22**. Furthermore, balance adjustment is performed between the power supply units **10** to **13** based on the control parameter Pr from the control terminal **20**.

(12) Each power supply unit **10** to **13** is a device with the same configuration, and is pre-associated with a unique identification number 0 to 3. The identification number is a unit identification information UID for the power supply units **10** to **13** to identify each other. The power supply units **10** to **13** are connected to each inter-unit communication line UCL, and communication is enabled between arbitrary power supply units **10** to **13** by using the unit identification information UID. Further, the power supply units **10** to **13** are connected to a pair of synchronous communication lines ZCL, thus enabling the output start timing to be synchronized.

(13) One of the power supply units **10** to **13** is determined beforehand to be the master unit MU, while the others are the slave units SU, and only the master unit MU is connected to the control terminal **20**. For example, the power supply unit **10** associated with the smallest identification number 0 is the master unit MU, and receives the control parameter Pr from the control terminal **20** via a terminal communication line TCL. However, the other power supply units **11** to **13** are the slave unit SU, and receive the control parameter Pr from the master unit MU via the inter-unit communication line UCL. Note that the master unit MU and the slave units SU indicate the functions of the power supply units **10** to **13** in data communication before the start of outputting, and the power supply units **10** to **13** after the start of the output operate without distinction between the master unit MU and the slave units SU.

(14) The control terminal **20** is a device that controls the power supply system **100**, generates a control parameter Pr of the power supply units **10** to **13**, and transmits the control parameter Pr to the master unit MU. The control parameter Pr is generated and outputted based on, for example, a user operation. For example, a PC on which a dedicated program is installed can be used as the control terminal **20**.

(15) The terminal communication line TCL is a wired or wireless communication path that is used for data communication between the control terminal **20** and the master unit MU. For communication with the control terminal **20**, for example, a packet communication system for transmitting and receiving TCP/IP (Transmission Control Protocol/Internet Protocol) packets can be adopted, and a connection can also be made via a Local Area Network (LAN) or the Internet.

(16) The inter-unit communication line UCL is a wired communication path that is used for data communication between the power supply units **10** to **13**. For the inter-unit communication, for example, a multi-master bus communication system as per the controller area network (CAN) standard is adopted, and data can be transmitted from any of the power supply units **10** to **13** to any of the power supply units **10** to **13**.

(17) The synchronous communication lines ZCL are wired communication paths for transmitting a trigger signal indicating the start of outputting, and each include a pair of signal lines isolated from the power supply units **10** to **13** and having a photocoupler interposed therebetween. The pair of

synchronous communication lines ZCL are capable of transitioning between a conductive state and a non-conductive state, and the trigger signal is a pulse signal that causes conduction between the pair of synchronous communication lines ZCL in a fixed time interval, and is transmitted from the master unit MU (power supply unit **10**) to all the slave units SU (power supply units **11** to **13**).

(18) (2) Power Supply Units **10** to **13**

(19) FIG. **2** is a diagram showing an example of a detailed configuration of the power supply units **10** to **13**. The power supply units **10** to **13** include a power supply main circuit **30**, a voltage detection unit **301**, a current detection unit **302**, a terminal communication unit **31**, an inter-unit communication unit **32**, a PWM drive unit **33**, a parameter storage unit **34**, a synchronous processing unit **35**, and a control unit **36**.

(20) A) Power Supply Main Circuit **30**

(21) The power supply main circuit **30** is a circuit that converts AC power inputted from the power grid via an input terminal  $P_i$  into DC power and outputs the DC power to the load **22** via an output terminal  $P_o$ . The operation of the power supply main circuit **30** is controlled based on the control parameter  $P_r$ . Specifically, when the operation mode is constant current operation (CC), the output current operates to match the target value  $C_r$ , and when the operation mode is constant voltage operation (CV), an output voltage  $V_o$  is controlled to match the target value  $C_r$ . The target value  $C_r$  is a target current  $I_{ref}$  in constant current operation or a target voltage  $V_{ref}$  in constant voltage operation.

(22) The power supply main circuit **30** is a bidirectional power supply (regenerative power supply) capable of bidirectionally converting AC power and DC power, and capable not only of performing a power operation to convert an AC voltage into a DC voltage and supply energy to the load **22**, but also of performing a regenerative operation to convert a DC voltage into an AC voltage and absorb energy from the load **22**, according to the status of the load **22**.

(23) The power supply main circuit **30** includes an AC/DC converter **40** that converts an AC voltage into a DC voltage, an isolation transformer **41** for isolating the input from the output, and a step-up/down DC/DC converter **42** that controls the output voltage  $V_o$ .

(24) The AC/DC converter **40** is a bidirectional converter capable of bidirectionally converting AC power and DC power, and can be configured using, for example, a three-state switching element.

(25) The isolation transformer **41** is a bidirectional DC/DC converter capable of bidirectionally transmitting DC power while ensuring isolation between the primary side and the secondary side, and includes a transformer **412**, a DC/AC converter **411** connected to the primary side of the transformer **412**, and an AC/DC converter **413** connected to the secondary side of the transformer **412**. The direct current inputted from the AC/DC converter **40** is converted into an alternating current by the DC/AC converter **411**, and, after passing through the transformer **412**, is converted into a direct current again by the AC/DC converter **413**. When the alternating current passing through the transformer **412** has a high frequency, isolation can be performed using a small transformer, and the entire device can be downsized.

(26) The step-up/down DC/DC converter **42** is a bidirectional DC/DC converter that controls the output voltage  $V_o$  based on the control parameter  $P_r$ , and a chopper circuit including a switching circuit **421** and an inductance **422**, for example, can be used. The switching circuit **421** includes two switching elements connected in series, and one end of the inductance **422** is connected to a connection point between the two switching elements. The two switching elements operate so as to alternately repeat a state in which one element is turned on while the other element is turned off, and a state in which one element is turned off while the other element is turned on, based on a PWM signal from the PWM drive unit **33**, and generate the output voltage  $V_o$  according to the corresponding duty ratio.

(27) B) Voltage Detection Unit **301** and Current Detection Unit **302**

(28) The voltage detection unit **301** is means for detecting the output voltage  $V_o$  of the power supply main circuit **30**. The detection value of the voltage detection unit **301** is inputted to the

control unit **36** as a detected voltage  $V_{det}$ . The current detection unit **302** is means for detecting an output current  $I_o$  of the power supply main circuit **30**. The detection value of the current detection unit **302** is inputted to the control unit **36** as a detected current  $I_{det}$ .

(29) C) Terminal Communication Unit **31**

(30) The terminal communication unit **31** is a packet communication means that communicates with the control terminal **20** via the terminal communication line TCL. The terminal communication unit **31** of the master unit MU receives the control parameter  $P_r$  from the control terminal **20** and outputs the control parameter  $P_r$  to the control unit **36**. The terminal communication unit **31** of the slave unit SU is not used, and the terminal communication unit **31** of the master unit MU acquires not only the control parameter  $P_r$  of the master unit MU but also the control parameter  $P_r$  of the slave unit SU.

(31) D) Inter-Unit Communication Unit **32**

(32) The inter-unit communication unit **32** is means for performing data communication between the power supply units **10** to **13** via the inter-unit communication line UCL. In inter-unit communication, any of the power supply units **10** to **13** is capable of outputting transmission data onto the inter-unit communication line UCL. The transmission data includes the unit identification information UID of the transmission source or the transmission target, and each power supply unit **10** to **13** monitors the transmission data on the inter-unit communication line UCL and selectively receives data necessary for that power supply unit based on the unit identification information UID of the transmission source or the transmission target.

(33) The control parameter  $P_r$  for the slave units SU is transmitted from the master unit MU to the slave units SU by inter-unit communication. The detected current  $I_{det}$  and the detected voltage  $V_{det}$  detected by each power supply unit **10** to **13** are transmitted as balance information  $B_i$  to the other power supply units **10** to **13** by inter-unit communication.

(34) E) PWM Drive Unit **33**

(35) The PWM drive unit **33** generates a PWM (Pulse Wide Modulation) signal based on a drive signal  $D_r$  generated by the control unit **36**, and outputs the PWM signal to the switching circuit **421**. Therefore, the power supply main circuit **30** outputs a voltage corresponding to the drive signal  $D_r$ .

(36) F) Parameter Storage Unit **34**

(37) The parameter storage unit **34** is storage means for storing various parameters referred to by the control unit **36**, and holds the control parameter  $P_r$  received from the control terminal **20**. The control parameter  $P_r$  includes operation mode information  $C_m$ , the target value  $C_r$ , and balance adjustment target information  $C_b$ .

(38) The operation mode information  $C_m$  is a parameter indicating constant current operation (CC) or constant voltage operation (CV) as the control method of the power supply system **100**, and is a parameter common to each power supply unit **10** to **13**. The target value  $C_r$  is a target current  $I_{ref}$  during constant current operation (CC) and a target voltage  $V_{ref}$  during constant voltage operation (CV). The balance adjustment target information  $C_b$  is an identification number of one or more other power supply units **10** to **13** to be referred to when balance adjustment is performed. The balance adjustment target information  $C_b$  is automatically generated by the user designating the unit configuration and the operation mode on the control terminal **20**. Therefore, the user is able to construct the power supply system **100** without being conscious of balance adjustment.

(39) G) Synchronous Processing Unit **35**

(40) The synchronous processing unit **35** is means for synchronizing the operation timing of the power supply units **10** to **13**, and transmits and receives a trigger signal TG for matching the operation timing via synchronous terminals **Z1** and **Z2**.

(41) The synchronous processing unit **35** is capable of outputting the trigger signal TG to the synchronous communication lines ZCL, and of detecting the trigger signal on the synchronous communication lines ZCL. The trigger signal TG is outputted as a pulse signal corresponding to a

conductive state of the pair of synchronous communication lines ZCL. Therefore, the trigger signal TG can be quickly detected in all the power supply units **10** to **13**, thereby yielding a high noise resistance and making it possible to suppress malfunction.

(42) When the trigger transmission signal sTG is inputted from the control unit **36**, the synchronous processing unit **35** of the master unit MU outputs the trigger signal TG to the pair of synchronous communication lines ZCL. However, the synchronous processing units **35** of the master unit MU and the slave units SU monitor the synchronous communication lines ZCL, and upon detecting the trigger signal TG, output a trigger reception signal rTG to the control unit **36**.

(43) H) Control Unit **36**

(44) The control unit **36** generates the drive signal Dr based on the detected voltage Vdet or the detected current Idet, and controls the power supply main circuit **30** via the PWM drive unit **33**. In addition, the terminal communication unit **31** and the inter-unit communication unit **32** are controlled to perform data communication with the control terminal **20** and other power supply units **10** to **13**. Furthermore, the synchronous processing unit **35** is controlled to perform synchronized control of the output start timing between the power supply units **10** to **13**.

(45) (3) Detailed Configuration of Synchronous Processing Unit **35**

(46) FIG. **3** is a diagram showing an example of a detailed configuration of the synchronous processing unit **35**. The synchronous processing unit **35** has two synchronous terminals Z1 and Z2, and the respective terminals Z1 and Z2 of the two or more power supply units **10** to **13** are connected using a pair of synchronous communication lines ZCL. The synchronous processing unit **35** transmits a trigger signal by establishing conduction between the pair of synchronous communication lines ZCL, and receives the trigger signal by detecting a conductive state between the synchronous communication lines ZCL.

(47) The synchronous processing unit **35** includes two photocouplers **51** and **52**. The photocoupler **51** is a transmission photocoupler for transmitting the trigger signal, and the photocoupler **52** is a reception photocoupler for receiving the trigger signal.

(48) The transmission photocoupler **51** includes a light-emitting element **511** driven by a trigger transmission signal sTG from the control unit **36** and a light-receiving element **512** that establishes conduction between the terminals Z1 and Z2 when the light-emitting element **511** emits light. When the trigger transmission signal sTG is not inputted, there is no conduction between the terminals Z1 and Z2, and when the trigger transmission signal sTG is inputted from the control unit **36**, conduction is established between the terminals Z1 and Z2.

(49) The reception photocoupler **52** includes a light-emitting element **521** driven by conduction between the pair of synchronous communication lines ZCL, and a light-receiving element **522** that outputs a trigger reception signal rTG when the light-emitting element **521** emits light. For this reason, the trigger reception signal rTG is not outputted when there is no conduction between the pair of synchronous communication lines ZCL, whereas the trigger reception signal rTG is outputted to the control unit **36** when there is conduction between the pair of synchronous communication lines ZCL.

(50) The terminals Z1 and Z2 of all the power supply units **10** to **13** are respectively connected via a pair of synchronous communication lines ZCL, and the light-receiving element **512** of the transmission photocoupler **51** is connected in parallel. Therefore, when the trigger signal is transmitted from any one of the power supply units **10** to **13**, the pair of synchronous communication lines ZCL conduct. The light-emitting element **521** of the reception photocoupler **52** is connected in series with the light-receiving element **512** of the transmission photocoupler **51**, detects a conductive state or a non-conductive state between the pair of synchronous communication lines ZCL, and generates a trigger reception signal rTG.

(51) FIG. **4** is a timing chart showing an example of an output start operation of the power supply system **100**. (a) in FIG. **4** represents the trigger transmission signal sTG in the master unit MU, and (b) and (c) are the trigger reception signal rTG and the drive signal Dr of each of the power supply



units **10** to **13**. Here, the operation mode is constant voltage (CV) operation, and output control is performed so that the output voltage  $V_o$  matches the target voltage  $V_{ref}$ . In addition, at the start of the output, soft start control is performed in which the output voltage  $V_o$  is changed at a predetermined time or rate of change so as to then reach the target voltage  $V_{ref}$ . Both the master unit MU and the slave units SU start to output power based on the trigger reception signal rTG.

(52) In the master unit MU, after the inter-unit communication unit **32** transmits the control parameter  $P_r$  to all the slave units SU, the control unit **36** generates the trigger transmission signal sTG, and the synchronous processing unit **35** outputs the trigger signal TG to the synchronous communication lines ZCL. The trigger transmission signal sTG is generated as a pulse signal having a predetermined width.

(53) In the master unit MU and the slave units SU, when the pair of synchronous communication lines ZCL conducts, the synchronous processing unit **35** generates the trigger reception signal rTG and outputs the trigger reception signal rTG to the control unit **36**. The control unit **36** starts outputting the drive signal  $D_r$  based on the timing of receiving the trigger reception signal rTG, and the power supply main circuit **30** starts outputting power. In FIG. 4, the drive signal  $D_r$  is outputted based on the leading edge of the trigger reception signal rTG, but the drive signal  $D_r$  can also be outputted based on the trailing edge of the trigger reception signal rTG.

(54) The trigger reception signal rTG is a pulse signal similar to the trigger transmission signal sTG, and is delayed by a delay time  $t_D$  after the trigger transmission signal sTG. The delay time  $t_D$  is mainly a delay time of the photocouplers **51** and **52**, and is on the order of several tens of microseconds. Therefore, the time is sufficiently shorter than the control cycle by the control unit **36**, for example, 250 microseconds, and can be ignored.

(55) Here, in a case where the output start command is transmitted between the power supply units **10** to **13** by digital communication via the inter-unit communication line UCL, the communication time becomes much longer than the delay time  $t_D$ , and the communication time varies. Therefore, there is a problem that a significant deviation occurs in the output start timing of the power supply units **10** to **13**, and the operation at the start of the output becomes unstable. In contrast, by quickly transmitting a trigger signal via the synchronous communication lines ZCL, the timing for starting to output power can be substantially synchronized, and the operation at the start of the output can be stabilized. In exactly the same manner, even in a case where the target value is changed during the output, the output change timing of the power supply units **10** to **13** can be substantially synchronized by transmitting the trigger signal via the isolated communication lines ZCL, and the operation at the time of the output change can be stabilized.

(56) Furthermore, by transmitting the trigger signal in a conductive or non-conductive state, noise resistance can be improved and malfunction can be prevented. Further, the noise resistance can be further improved by interposing a photocoupler.

(57) Note that, because the delay time  $t_D$  of the photocouplers **51** and **52** is an extremely short time, the master unit MU is also capable of outputting the drive signal  $D_r$  based on the trigger transmission signal sTG instead of the trigger reception signal rTG.

(58) (4) Operation at Start of Output

(59) Steps **S101** to **S106** in FIG. 5 are a flowchart showing an example of the operation of the master unit MU. This flowchart is started by a data transmission request from the control terminal **20**.

(60) The terminal communication unit **31** receives the control parameter  $P_r$  from the control terminal **20** via the terminal communication line TCL (step **S101**). Among the received data, the control parameter  $P_r$  for the master unit is stored in the parameter storage unit **34**, and the inter-unit communication unit **32** transmits the control parameter  $P_r$  for the slave units to each slave unit SU via the inter-unit communication line UCL (step **S102**). Each slave unit SU, having received the control parameter  $P_r$ , transmits a data reception confirmation reply to the master unit MU.

(61) The inter-unit communication unit **32** receives the confirmation replies from each slave unit

SU. After receiving a confirmation reply from all the slave units SU, the synchronous processing unit **35** transmits the trigger signal TG to the synchronous communication lines ZCL (steps **S103** and **S104**). When the synchronous processing unit **35** receives the trigger signal TG, the control unit **36** generates the drive signal Dr, and the power supply main circuit **30** starts outputting (steps **S105** and **S106**).

(62) Steps **S201** to **S204** in FIG. **6** are a flowchart showing an example of the operation of the slave units SU. This flowchart is started by transmission of the control parameter Pr from the master unit MU.

(63) The inter-unit communication unit **32** receives the control parameter Pr from the master unit MU (step **S201**). The received control parameter Pr is stored in the parameter storage unit **34**, and a reception confirmation reply is transmitted to the master unit MU (step **S202**). Thereafter, when the synchronous processing unit **35** receives the trigger signal TG, the control unit **36** generates the drive signal Dr, and the power supply main circuit **30** starts outputting (steps **S203** and **S204**).

#### Second Embodiment

(64) In the above embodiment, an example of a power supply system **100** that transmits a trigger signal via the synchronous communication lines ZCL and synchronizes the output start timing of the power supply units **10** to **13** has been described. In contrast, in the present embodiment, an example in which the occurrence of an abnormality in any power supply unit **10** to **13** is quickly transmitted to the other power supply units **10** to **13** by using the synchronous communication lines ZCL will be described.

(65) When the power supply system **100** is supplying power to the load **22**, in a case where failure occurs in any of the two or more power supply units **10** to **13**, for example in a case where an abnormal temperature rise is detected, an excessive load is applied to the other power supply units **10** to **13** due to a reduction in the output of the power supply unit **10** to **13**, and there is a risk of damage to the normal power supply units **10** to **13**. Therefore, in a case where an abnormality is found in any of the power supply units **10** to **13** that are outputting power, it is desirable to quickly stop the output operation of all the power supply units **10** to **13**. Therefore, by using the synchronous communication lines ZCL to transmit the error signal ER from the failed power supply unit **10** to **13** to the normal power supply units **10** to **13**, the outputs of all the power supply units **10** to **13** are quickly stopped. The error signal ER is transmitted to the synchronous signal lines ZCL as a signal that is distinguishable from the trigger signal TG to start outputting.

(66) FIG. **7** is a timing chart showing an example of the operation of the power supply system **100** according to the second embodiment of the present invention. (a) to (d) in FIG. **7** show the trigger transmission signals sTG inputted to the synchronous processing unit **35** of the power supply units **10** to **13**, and (e) shows the drive signal Dr of the power supply units **10** to **13**. Because the configuration of the power supply system **100** is the same as that in the case of FIG. **1**, redundant descriptions will be omitted.

(67) The trigger signal TG is a signal transmitted only by the master unit MU, whereas the error signal ER is a signal that is transmittable by any power supply unit **10** to **13**, regardless of whether same is the master unit MU or a slave unit SU. In addition, the trigger signal TG is a signal for starting outputting and is transmitted before power output is started, whereas the error signal ER is a signal providing an alert regarding the occurrence of an abnormality and for stopping the output, which is transmitted during the power output.

(68) The trigger signal TG and the error signal ER are distinguished by affording the signals different signal output times. For example, the pulse width wER of the error signal ER is longer than the pulse width wTG of the trigger signal TG, and in a case where any signal is outputted on the synchronous communication lines ZCL, it is possible to discriminate, based on the signal duration, whether the signal is the trigger signal TG or the error signal ER.

(69) In a case where the master unit MU (UID=0) outputs the trigger signal TG, each slave unit SU (UID=1 to 3) is capable of discriminating that the signal is the trigger signal TG because the signal

duration is  $wTG$ . The control unit **36** of each power supply unit **10** to **13** generates the drive signal  $Dr$  based on the timing of the trailing edge of the trigger reception signal  $rTG$ , and the power supply main circuit **30** starts outputting.

(70) When failure occurs in a slave unit  $SU$  ( $UID=2$ ) that is outputting power, the error signal  $ER$  is outputted from the power supply unit **13**. Because the duration of this signal is  $wER$ , the signal can be discriminated as being the error signal  $ER$ . The control unit **36** of each power supply unit **10** to **13** stops the generation of the drive signal  $Dr$  based on the timing of the trailing edge of the error signal  $ER$ , and the power supply main circuit **30** stops outputting.

(71) FIG. **8** is a timing chart showing another example of the operation of the power supply system **100** according to the second embodiment of the present invention. (a) to (d) in FIG. **8** show the trigger transmission signals  $sTG$  inputted to the synchronous processing unit **35** of the power supply units **10** to **13**, and (e) shows the drive signal  $Dr$  of the power supply units **10** to **13**.

(72) The trigger signal  $TG$  and the error signal  $ER$  are distinguished from each other by affording same different numbers of times the pulse signals are continuously outputted. For example, although the widths of the pulse signals constituting the trigger signal  $TG$  and the error signal  $ER$  are the same, the trigger signal  $TG$  is a single pulse transmitted at sufficient time intervals, whereas the error signal  $ER$  is continuously outputted with two pulse signals sandwiching a short interval time  $tB$ .

(73) The power supply units **10** to **13** which receive the trigger signal  $TG$  and the error signal  $ER$  are capable of discriminating, after detecting one pulse signal, and after the interval time  $tB$  has elapsed, whether the signal is the trigger signal  $TG$  or the error signal  $ER$ .

(74) According to the present embodiment, not only the trigger signal  $TG$  but also the error signal  $ER$  can also be transmitted via the synchronous communication lines  $ZCL$ . Therefore, in a case where an abnormality such as failure occurs in any of the power supply units **10** to **13**, the outputs of all the power supply units can be quickly stopped, and damage to the normal power supply units can be prevented. Moreover, the error signal  $ER$  can be realized at low cost by being transmitted between the power supply units **10** to **13** as a signal that can be distinguished from the trigger signal  $TG$ .

## DESCRIPTION OF REFERENCE NUMERALS

(75) **100** power supply system **10** to **13** power supply unit **20** control terminal **21** AC power **22** load **30** power supply main circuit **301** voltage detection unit **302** current detection unit **31** terminal communication unit **32** inter-unit communication unit **33** PWM drive unit **34** parameter storage unit **35** synchronous processing unit **36** control unit **41** isolation transformer **412** transformer **421** switching circuit **422** inductance **42** step-up/down DC/DC converter **51** transmission photocoupler **52** reception photocoupler  $Bi$  balance information  $Pr$  control parameter  $Cm$  operation mode information  $Cr$  target value  $Cb$  balance adjustment target information  $Dr$  drive signal  $Idet$  detected current  $Io$  output current  $Iref$  target current  $Vdet$  detected voltage  $Vo$  output voltage  $Vref$  target voltage  $MU$  master unit  $SU$  slave unit  $TCL$  terminal communication line  $UCL$  inter-unit communication line  $ZCL$  synchronous communication line  $TG$  trigger signal  $rTG$  trigger reception signal  $sTG$  trigger transmission signal  $ER$  error signal

## Claims

1. A power supply system comprising two or more power supply units, the power supply units cooperating to supply DC power to a shared load, wherein each of the power supply units includes: a power supply main circuit that converts AC power inputted from the outside into DC power; a control unit that controls the power supply main circuit; a pair of synchronous terminals respectively connected to a pair of synchronous signal lines that transmit an error signal to the other of the two or more power supply units; a transmission unit that outputs the error signal by establishing conduction between the pair of synchronous signal lines upon detection of an

abnormality; and a reception unit that detects a conductive state between the pair of synchronous signal lines and that receives the error signal, wherein the power supply main circuit stops outputting based on timing of receiving the error signal.

2. The power supply system according to claim 1, wherein the transmission unit and the reception unit each include a photocoupler, and wherein the pair of synchronous signal lines is isolated from the power supply units by the photocoupler.

3. The power supply system according to claim 1, wherein the power supply main circuit starts outputting based on an edge of the error signal.

4. A power supply unit that is connected to another power supply unit and that cooperates with the another power supply unit to supply DC power to a shared load, the power supply unit comprising: a power supply main circuit that converts AC power inputted from the outside into DC power; a control unit that controls the power supply main circuit; a pair of synchronous terminals respectively connected to a pair of synchronous signal lines that transmit an error signal to the another power supply unit; a transmission unit that outputs the error signal by establishing conduction between the pair of synchronous signal lines upon detection of an abnormality; and a reception unit that detects a conductive state between the pair of synchronous signal lines and that receives the error signal, wherein the power supply main circuit stops outputting based on timing of receiving the error signal.

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